Uniform stability of approximately idempotent laws on finite groups*

Sam Vaseghi†

Wisehouse Science Communication Lab, Wisehouse AB

28 October 2026

Abstract

For every $C > 2$, a probability law on a finite group with sufficiently small self-convolution defect is within C times that defect of the uniform law on a subgroup, in full ℓ^1 norm. The threshold depends only on C, independently of the group, and no symmetry or commutativity assumption is required. The coefficient 2 cannot hold at any universal positive threshold. The proof obtains approximate stochastic factors from the corrected small-projection theorem of Zippin, then uses translation invariance to recover an exact permutation action and a partition into subgroup cosets. A local estimate converts this coarse approximation into the linear bound. We give an explicit sufficient threshold for $C = 4$ and separate the nonabelian reconstruction from the abelian consequence of Saeki's norm-gap theorem.

1 Introduction and statement

If two independent samples with law μ are multiplied in a finite group, their product has law $\mu * \mu$. Exact invariance under this operation characterizes the uniform laws on subgroups, by the Kawada–Itô theorem [1]. We consider whether approximate invariance gives a quantitative version of that characterization, uniformly over all finite groups.

For a finite group G, convolution and norm are

$$(\lambda * \rho)(z) = \sum_{xy=z} \lambda(x)\rho(y), \qquad \|\lambda\|_1 = \sum_{x\in G} |\lambda(x)|.$$

Thus our distance is twice probabilistic total variation. We also write $\lambda\rho$ for convolution and $m_H = \mathbf{1}_H/|H|$ for the uniform law on a subgroup $H \le G$.

Theorem 1.1. *For every real $C > 2$, there is $\delta_C > 0$ with the following property. For every finite group G and every probability law μ on G, if*

$$\delta = \|\mu * \mu - \mu\|_1 \le \delta_C,$$

then some subgroup $H \le G$ satisfies

$$\|\mu - m_H\|_1 \le C\delta.$$

The threshold depends only on C. In particular, the conclusion holds with $C = 4$ and an absolute positive threshold.

*An earlier unrefereed version is available at https://doi.org/10.5281/zenodo.23081071.

†Email: sam.vaseghi@wisehouse-publishing.com.

Proposition 4.3 shows that the coefficient 2 cannot replace $C > 2$ at any universal positive threshold. Both assertions are uniform over the group: the threshold in the theorem is independent of $|G|$, while the counterexample uses groups of increasing order. No symmetry assumption is imposed on μ, and the subgroup H need not be normal. Thus the conclusion identifies the structure of a law from its convolution defect without fixing the ambient group in advance.

The approximate finite-group question was posed publicly by Ekene E. [2]. In his answer, Freddie Manners described the strategy of first obtaining a uniform coarse approximation by a subgroup law and then improving it locally to a linear estimate; the proposed coarse approximation lemma was left unproved there. We use that strategy. The local calculation in Lemma 4.1 makes the limiting coefficient 2 explicit. The theorem does not establish the full defect range $\delta \le 1/10$ asked about in the public question, or provide an efficient sampling algorithm.

In the abelian case, a uniform approximation already follows from classical idempotent-measure theory. Saeki's norm gap [3], in the form restated by Mudge–Pham [4, Theorem 1.1(i)], combines with the spectral repair in Lemma 2.2 to give this conclusion. Indeed, that repair produces a real convolution idempotent ν of mass one with

$$\|\nu - \mu\|_1 \le 4\delta, \qquad \|\nu\|_1 \le 1 + 4\delta.$$

For sufficiently small δ, the latter norm is less than $(1 + \sqrt{2})/2$. On the dual group, Saeki's theorem says that the Fourier transform of ν is the indicator of a coset. Since the transform equals one at the trivial character, this coset is a subgroup, and ν is the uniform law on its annihilator. Lemma 4.1 then gives every coefficient $C > 2$. This abelian deduction is a baseline for the result here; a theorem about pointwise idempotent Fourier multipliers on a nonabelian group is not directly a classification of convolution idempotents on that group.

For general finite groups, the main task is to obtain the coarse subgroup approximation. Algebraic repair of a nearly idempotent convolution operator gives an exact projection, but that projection may have negative entries and need not itself average over a subgroup. Zippin's corrected small-projection theorem [5, 6], also stated in his later exposition [7, Theorem 2.3], supplies a different way to use the repaired projection. With the quantitative bound recorded by Godefroy–Kalton–Lancien [8, p. 809, proof of Fact 3.3], it gives stochastic factors through a finite set of labels, with an error tending to zero independently of the number of labels or group elements.

The factors initially have no exact compatibility with group translations. In Section 3, we show that translations nevertheless induce maps on the labels that are close to permutations. The separation between distinct permutations makes the resulting action exact. Averaging the factors over this action then produces a partition of G into cosets, and hence the required subgroup law. Section 4 uses the local estimate to replace the coarse error by $C\delta$. In particular, the small-projection theorem supplies a vanishing error; the linear dependence on δ comes from the final local argument.

A stronger, linear approximate stochastic retract statement was previously formulated by Osborne [9, §4, equations (F2)–(F3)]. Appendix A records a derivation from the scalar consequences of Kitaev's approximate-algebra theorem [10]. We do not claim that factorization statement as a new result. The additional deduction developed here is the recovery of the subgroup from approximate stochastic factors of convolution operators, together with the uniform conclusion and the failure of the endpoint coefficient 2.

2 Approximate stochastic factors from small projections

Every matrix acts on column vectors of functions. Its norm is the operator norm for the supremum norms, hence the maximum absolute row sum, including for rectangular matrices.

A stochastic matrix has nonnegative entries and every row sums to one. We write $\mathbf{1}$ for the constant vector of the required size; a map is unital when it preserves this vector.

The factors sought below describe an approximate averaging operation. For an exact model, partition a finite set into nonempty cells $C_1, \dots, C_k$ and define

$$A_{xj} = \mathbf{1}_{C_j}(x), \qquad M_{jx} = \frac{\mathbf{1}_{C_j}(x)}{|C_j|}.$$

Then A assigns a cell value to each point, M averages a function over each cell, and $MA = I_k$. Their product AM replaces a function by its average on each cell. For a general nearly idempotent stochastic matrix, we first obtain stochastic A, M for which AM approximates the given matrix and MA approximates the identity. These factors need not yet come from a partition. Section 3 recovers such a partition when the matrix is a convolution operator.

The construction begins with algebraic repair of idempotence. The repaired projection may lose positivity, so we also need the following estimate for replacing a nearly positive row by a probability row.

Lemma 2.1 (Repairing a real row)**.** *Let v be a real row with sum $s > 0$, positive mass t, negative mass N, and absolute mass $\ell = t + N$. Then $w = v_+/t$ is a probability row and*

$$\|v - w\|_1 = N + |t - 1| = \max\{\ell - 1, 1 - s\}.$$

In particular, if $\ell \le 1 + \alpha$ and $|s - 1| \le \beta < 1$, this distance is at most $\max\{\alpha, \beta\}$.

Proof. The negative coordinates contribute N to the distance; the positive coordinates contribute $|t - 1|$. Substituting $s = t - N$ proves the identity, separately for $t \ge 1$ and $t \le 1$. □

Lemma 2.2 (Spectral repair)**.** *Let Q be a stochastic $n \times n$ matrix, $n \ge 1$, and let $\eta = \|Q^2 - Q\| \le 1/100$. There is a real projection P such that $P\mathbf{1} = \mathbf{1}$ and*

$$d := \|P - Q\| \le 4\eta, \qquad \|P\| \le 1 + d.$$

Proof. The convergent binomial series defines

$$P = \frac{1}{2}\left[I + (2Q - I)(I + 4(Q^2 - Q))^{-1/2}\right]. \tag{2.1}$$

All factors commute, and $(2Q - I)^2 = I + 4(Q^2 - Q)$, so $P^2 = P$. The coefficients are real, and $P\mathbf{1} = \mathbf{1}$. Since $\|2Q - I\| \le 3$, the absolute binomial series gives

$$d \le \frac{3}{2}\left((1 - 4\eta)^{-1/2} - 1\right) \le 4\eta. \tag{2.2}$$

For the final inequality, the derivative of the preceding expression is $3(1 - 4\eta)^{-3/2} < 4$ on $[0, 1/100]$. Positivity of P is not asserted. □

For clarity, we state the Banach-space input in the form used below. A real Banach space E is a P_λ space if every isometric copy of E in a Banach space is the range of a projection of norm at most λ. For spaces of the same finite dimension, the Banach–Mazur distance is

$$d_{\mathrm{BM}}(E, F) = \inf_T \|T\| \left\|T^{-1}\right\|,$$

where the infimum is over linear isomorphisms $T\colon F \to E$. The corrected small-projection theorem gives, for a finite-dimensional P_{1+d} space and $v = d^{1/8} < 1/17$,

$$d_{\mathrm{BM}}(E, \ell_\infty^{\dim E}) \le D(d) := \frac{1 + 6v}{(1 - 6v)(1 - 17v)}. \tag{2.3}$$

The displayed bound is recorded in [8, p. 809], with its attribution to Zippin and to Bernau's Theorem 13. In particular, the distance tends to one as d tends to zero, independently of the dimension.

Lemma 2.3 (Coarse stochastic factorization)**.** *There are an absolute $\eta_0 > 0$ and a nondecreasing function $\omega\colon [0,\eta_0] \to [0,\infty)$, with $\omega(0) = 0$ and $\omega(\eta) \to 0$ as $\eta \downarrow 0$, such that every stochastic matrix Q with $\eta = \|Q^2 - Q\| \le \eta_0$ has stochastic factors*

$$A\colon \mathbb{R}^k \to \mathbb{R}^n, \qquad M\colon \mathbb{R}^n \to \mathbb{R}^k$$

for some $k \ge 1$, satisfying

$$\|AM - Q\| \le \omega(\eta), \qquad \|MA - I_k\| \le \omega(\eta). \tag{2.4}$$

For sufficiently small η, one may take

$$\beta(\eta) = D(4\eta) - 1, \qquad \omega(\eta) = \frac{2(4\eta + \beta(\eta))}{1 - \beta(\eta)}. \tag{2.5}$$

Proof. Let P be the real projection from Lemma 2.2, and put $E = \operatorname{im} P$ over $\mathbb{R}$. Since $P\mathbf{1} = \mathbf{1}$, its rank k is positive. The space E is $P_{1+4\eta}$. Indeed, for an isometric inclusion $i\colon E \to Y$, extend each coordinate of $i^{-1}\colon i(E) \to \ell_\infty^n$ by Hahn–Banach. This gives $J\colon Y \to \ell_\infty^n$ of norm at most one, and iPJ is a projection onto $i(E)$ of norm at most $1 + 4\eta$.

By (2.3), there is a real isomorphism $T_0\colon \ell_\infty^k \to E$ such that

$$\|T_0\| \le 1, \qquad \left\|T_0^{-1}\right\| \le 1 + \beta, \qquad \beta = D(4\eta) - 1.$$

Here the Banach–Mazur infimum is attained: normalize an approximating sequence to norm one and take a convergent subsequence in the finite-dimensional space of operators; a uniform inverse bound makes its limit invertible. When $\eta = 0$, apply (2.3) with any positive excess tending to zero and use the same attainment argument to obtain $\beta = 0$.

We next arrange that the isomorphism preserves the constant function. Put $u = T_0^{-1}\mathbf{1}$. For each coordinate j, $\|T_0 e_j\| \ge 1/(1+\beta)$, so some row has $a = |(T_0)_{ij}| \ge 1/(1+\beta)$. Its remaining absolute mass is at most $1 - a$. Since $(T_0 u)_i = 1$ and $\|u\| \le 1 + \beta$,

$$1 \le a|u_j| + (1 - a)(1 + \beta), \qquad |u_j| \ge 1 + \beta - \frac{\beta}{a} \ge 1 - \beta^2.$$

Choose $\eta_0 \le 1/100$ small enough that $(4\eta_0)^{1/8} < 1/17$ and $\beta(\eta_0) < 1$. Then

$$T = T_0 \operatorname{diag}(u), \qquad S = T^{-1}P$$

are real and exactly unital, $ST = I_k$, $TS = P$, and

$$\|T\| \le 1 + \beta, \qquad \left\|T^{-1}\right\| \le \frac{1}{1 - \beta}, \qquad \|S\| \le \frac{1 + d}{1 - \beta}.$$

Apply Lemma 2.1 to the rows of T and S. The resulting stochastic matrices A, M satisfy

$$\|A - T\| \le \beta, \qquad \|M - S\| \le \sigma := \frac{d + \beta}{1 - \beta}.$$

Using $\|A\| = \|M\| = 1$ gives

$$\|AM - Q\| \le d + \beta + (1 + \beta)\sigma = \frac{2(d + \beta)}{1 - \beta},$$
$$\|MA - I_k\| \le \beta + (1 + \beta)\sigma \le \frac{2(d + \beta)}{1 - \beta}.$$

Since $d \le 4\eta$, this proves (2.5). The expression is nondecreasing on a sufficiently small interval and tends to zero with η, with value zero at $\eta = 0$. □

The modulus in this proof is coarse: it is of order $\eta^{1/8}$. The next two sections explain why a linear modulus at this stage is unnecessary.

3 Recovering the group action and the subgroup

For a probability law μ on G, define right-convolution on functions by

$$(Q_\mu f)(x) = \sum_h \mu(h) f(xh).$$

The same definition applies to signed functions on G. It gives $Q_\lambda Q_\rho = Q_{\lambda * \rho}$ and $\|Q_\lambda\| = \|\lambda\|_1$: each row of Q_λ is a permutation of the values of λ. In particular, the operator defect of Q_μ equals the convolution defect δ. Let

$$(R_g f)(x) = f(g^{-1}x).$$

These matrices satisfy $R_g R_h = R_{gh}$ and commute with every Q_μ. In the exact coset-averaging model, left translation of G permutes the cells. Our next task is to recover this permutation action from the approximate factors. The elementary fact we need is that two stochastic maps which nearly invert one another must be close to permutations.

Lemma 3.1 (Almost stochastic inverses)**.** *If B, C are stochastic $k \times k$ matrices and*

$$\|BC - I\|, \|CB - I\| \le 2r, \qquad 0 \le r < 1/2,$$

there is a permutation matrix V such that

$$\|B - V\| \le \frac{2r}{1-r}.$$

Proof. A stochastic row is at distance $2(1-d_i)$ from the ith unit row when its ith entry is d_i. Thus $(BC)_{ii} \ge 1 - r$. For each i choose $j = j(i)$ with $C_{ji} \ge 1 - r$, using the convex average $(BC)_{ii}$. From $(CB)_{jj} \ge 1 - r$ we get

$$1 - r \le C_{ji}B_{ij} + 1 - C_{ji}, \qquad B_{ij} \ge 1 - \frac{r}{1-r}.$$

The selected $j(i)$ are distinct, since otherwise a row of C would contain two entries at least $1 - r > 1/2$. Put the ones of V at $(i, j(i))$ and use the same row-distance identity. □

Proposition 3.2 (Equivariant reconstruction)**.** *Let $Q = Q_\mu$ on a finite group G. Suppose stochastic $A\colon \mathbb{R}^k \to \mathbb{R}^G$ and $M\colon \mathbb{R}^G \to \mathbb{R}^k$ satisfy*

$$\|AM - Q\| \le e, \qquad \|MA - I_k\| \le e, \qquad 0 \le e \le 1/100.$$

Then some $H \le G$ satisfies

$$\|Q - Q_{m_H}\| \le 50e.$$

Proof. The factorization errors first give

$$\|QA - A\| \le 2e, \qquad \|MQ - M\| \le 2e. \tag{3.1}$$

For every $g \in G$, put $B_g = MR_gA$. This map takes a function on the labels to a function on G, translates it, and returns it to the labels. It is stochastic because each factor is stochastic. Replacing AM by Q and then using commutation and (3.1) yields

$$\|B_g B_h - B_{gh}\| \le 3e, \qquad \|B_1 - I_k\| \le e. \tag{3.2}$$

For instance,

$$B_g B_h - B_{gh} = MR_g(AM - Q)R_h A + MR_{gh}(QA - A).$$

Both inverse products $B_gB_{g^{-1}}$ and $B_{g^{-1}}B_g$ are therefore within $4e$ of the identity. Lemma 3.1, with $r = 2e$, gives permutation matrices P_g with

$$\|B_g - P_g\| \le b := \frac{4e}{1-2e}.$$

Distinct permutation matrices have distance two. The approximate multiplication law gives

$$\|P_gP_h - P_{gh}\| \le \|P_gP_h - B_gB_h\| + \|B_gB_h - B_{gh}\| + \|B_{gh} - P_{gh}\| \le 3b + 3e < 2.$$

Hence $P_gP_h = P_{gh}$. In particular, P_1 is an idempotent permutation and therefore the identity. We have obtained an exact G-action on the k labels, using the pullback convention $(P_gv)(j) = v(g^{-1}j)$.

We next make both stochastic factors equivariant. Directly,

$$\|R_gA - AB_g\| \le 3e, \qquad \|MR_g - B_gM\| \le 3e.$$

For example, $R_gA - AMR_gA = R_g(A - QA) + (Q - AM)R_gA$. The other estimate uses $MQ - M$. Put

$$a = 3e + b, \qquad \bar{A} = \frac{1}{|G|}\sum_g R_gAP_g^{-1}, \qquad \bar{M} = \frac{1}{|G|}\sum_g P_gMR_g^{-1}.$$

Reindexing each average using the exact action shows that the averaged matrices are equivariant. They remain stochastic, and the preceding estimates give

$$R_g\bar{A} = \bar{A}P_g, \qquad P_g\bar{M} = \bar{M}R_g,$$
$$\left\|\bar{A} - A\right\|, \left\|\bar{M} - M\right\| \le a, \qquad \left\|\bar{M}\bar{A} - I_k\right\| \le e + 2a.$$

Writing $s = a + e/2$, every diagonal entry of $\bar{M}\bar{A}$ is at least $1 - s$, by the stochastic row-distance identity used in Lemma 3.1. The assumption $e \le 1/100$ gives $s \le 743/9800 < 1/2$. We use this diagonal concentration first to prove transitivity of the label action, and then to identify one cell for each label.

Take any label orbit O. Since $P_g\mathbf{1}_O = \mathbf{1}_O$ and $R_g\bar{A} = \bar{A}P_g$, the function $\bar{A}\mathbf{1}_O$ is invariant under every left translation of G. It is therefore constant, say c_O. For $j \in O$,

$$c_O = \sum_x \bar{M}_{jx}(\bar{A}\mathbf{1}_O)(x) \ge (\bar{M}\bar{A})_{jj} \ge 1 - s.$$

The constants c_O sum to one. Since $s < 1/2$, there cannot be two label orbits, so the action is transitive.

For a label j, let H_j be its stabilizer. Transitivity gives $|H_j| = |G|/k$. Equivariance says

$$\bar{A}_{gx,gj} = \bar{A}_{xj}, \qquad \bar{M}_{gj,gx} = \bar{M}_{jx}.$$

Consequently column j of $\bar{A}$ and row j of $\bar{M}$ are constant on each orbit H_jx. Transitivity on the labels and equivariance make all column sums of $\bar{A}$ equal. Their total is $|G|$, since $\bar{A}$ is stochastic, so each column sum is $|G|/k$. Each orbit H_jx also has $|G|/k$ elements. If α_O denotes the value of column j on such an orbit O, then

$$\frac{|G|}{k} = \sum_x \bar{A}_{xj} = \frac{|G|}{k}\sum_{O \in H_j\backslash G} \alpha_O, \qquad \sum_{O \in H_j\backslash G} \alpha_O = 1.$$

Thus the column values on the stabilizer orbits form a probability vector.

Since $\sum_x \bar{M}_{jx}\bar{A}_{xj} \geq 1-s$, some orbit C_j has column value at least $1-s$. This orbit is unique because $s < 1/2$. The choice is equivariant: $C_{gj} = gC_j$. Distinct chosen orbits are disjoint, since an element in $C_i \cap C_j$ would give two entries of one stochastic row at least $1-s$. There are k cells each of size $|G|/k$, so they partition G.

Outside C_j, column j is at most s. If $q_j = \bar{M}_j(C_j)$, then

$$1 - s \leq \sum_x \bar{M}_{jx}\bar{A}_{xj} \leq q_j + s(1-q_j), \qquad 1 - q_j \leq \frac{s}{1-s}.$$

The row $\bar{M}_j$ is constant on C_j. Define the exact partition factors by

$$(A_0)_{xj} = \mathbf{1}_{C_j}(x), \qquad (M_0)_{jx} = \frac{\mathbf{1}_{C_j}(x)}{|C_j|}.$$

For a point $x \in C_j$, row x of $\bar{A}$ assigns mass at least $1-s$ to label j, so its distance from the corresponding row of A_0 is at most $2s$. Similarly, the distance of $\bar{M}_j$ from the uniform row on C_j is $2(1-q_j)$. Therefore

$$\left\|\bar{A} - A_0\right\| \leq 2s, \qquad \left\|\bar{M} - M_0\right\| \leq \frac{2s}{1-s}.$$

An equivariant partition of G under left translation consists of left cosets of a subgroup. Indeed, let H be the cell containing the identity. If $h \in H$, then hH is a cell containing h, so $hH = H$. This gives closure and inverses, and every cell is xH. If j_0 labels H and $j = xj_0$, then $H_j = xHx^{-1}$ and the earlier orbit $H_j x$ equals xH. Thus the argument does not require H to be normal. Therefore $A_0 M_0 = Q_{m_H}$, in our right-convolution convention. Finally,

$$\|Q - Q_{m_H}\| \leq e + 2a + 2s + \frac{2s}{1-s} \leq 50e.$$

For the last estimate, $b/e \leq 200/49$, $a/e \leq 347/49$, and $s/e \leq 743/98$ when $e > 0$; substitute $e \leq 1/100$. The zero-error case follows from the same construction with $a = s = 0$. □

4 Local improvement and the limiting coefficient

The reconstruction proposition gives a subgroup law at a distance tending uniformly to zero with the defect. The following estimate, implementing the local strategy in [2], converts that conclusion into the claimed linear bound.

Lemma 4.1 (Local improvement)**.** *For any probability law μ and any $H \leq G$, put*

$$\delta = \left\|\mu^2 - \mu\right\|_1, \qquad t = \|\mu - m_H\|_1.$$

Then

$$\delta \geq \frac{t}{2} - t^2. \tag{4.1}$$

In particular, $t \leq 1/4$ implies $t \leq 4\delta$.

Proof. Write $q = m_H$, $\rho = \mu - q$, and $p = \mu(G \setminus H)$. The linear part of the defect at q is

$$L\rho = q\rho + \rho q - \rho.$$

For $x \in H$, both products $h^{-1}x$ and xh^{-1} lie in H as h ranges over H. Thus $(q\rho)(x) = (\rho q)(x) = -p/|H|$, because the mass of ρ on H is $-p$. Consequently

$$(L\rho)|_H = -\rho|_H - 2pq.$$

The mass of this restriction is $-p$. Since $L\rho$ has total mass zero, its restriction outside H has mass p. Therefore, with

$$b = \|(L\rho)|_H\|_1\,, \qquad c = \left\|(L\rho)|_{G\setminus H}\right\|_1\,,$$

we have $b, c \geq p$. On H, the identity above gives $\|\rho|_H\|_1 \leq b + 2p$, while outside H the signed function ρ equals the nonnegative law μ. Hence

$$t = \|\rho|_H\|_1 + p \leq b + 3p \leq 2(b + c) = 2\,\|L\rho\|_1\,.$$

Finally, $\mu^2 - \mu = L\rho + \rho^2$ and $\|\rho^2\|_1 \leq t^2$. Hence $t \leq 2(\delta + t^2)$, which is (4.1). The quadratic term includes products of elements outside H that return to H. □

Proof of Theorem 1.1. For fixed $C > 2$, choose $0 < \delta_C \leq \eta_0$ so small that

$$\omega(\delta_C) \leq \frac{1}{100}, \qquad 50\omega(\delta_C) \leq \frac{1}{2} - \frac{1}{C}.$$

This is possible by Lemma 2.3, independently of all dimensions. For $\delta \leq \delta_C$, apply that lemma to Q_μ and then Proposition 3.2 with $e = \omega(\delta)$. The resulting subgroup satisfies

$$t = \|\mu - m_H\|_1 \leq 50\omega(\delta) \leq \frac{1}{2} - \frac{1}{C}.$$

Lemma 4.1 gives $\delta \geq t(1/2 - t) \geq t/C$. This also includes $\delta = 0$. □

The classical estimates allow an explicit threshold, although they make no claim to a useful numerical range.

Corollary 4.2. *The conclusion of Theorem 1.1 holds with $C = 4$ and*

$$\delta_4 = \frac{1}{4 \cdot 12800^8}.$$

Proof. For $0 < \delta \leq \delta_4$, put $v = (4\delta)^{1/8} \leq 1/12800$. From (2.3),

$$\beta = \frac{29v - 102v^2}{1 - 23v + 102v^2} \leq 30v.$$

The denominator is positive, and the last inequality is equivalent to $1 - 588v + 3060v^2 \geq 0$, which holds on this interval. In particular, $v < 1/17$, $\beta < 1$, and $\delta < 1/100$, so all smallness conditions in the proof of Lemma 2.3 hold throughout this interval. Its construction gives factors with error

$$e \leq \frac{2(v^8 + 30v)}{1 - 30v} \leq 64v \leq \frac{1}{200}.$$

The middle inequality follows from $2v^7 + 1920v \leq 4$. Proposition 3.2 gives $t \leq 3200v \leq 1/4$, and Lemma 4.1 gives $t \leq 4\delta$. The zero-defect case follows from the theorem. □

Proposition 4.3 (The coefficient two cannot hold uniformly)**.** *There is no $\delta_* > 0$ such that every finite-group probability law with defect at most δ_* is within 2δ of some subgroup-uniform law.*

Proof. Let H have order $n \geq 3$, let $G = H \times C_2$, identify H with its index-two subgroup, and let u be uniform on the other coset. Write $q = m_H$, and for $0 < p < 1/2$ put

$$\mu = (1 - 2p)q + p\delta_1 + pu.$$

The relations $q^2 = u^2 = q$ and $qu = uq = u$ give

$$\mu^2 - \mu = p^2 q - p(1-p)\delta_1 + p(1-2p)u.$$

Reading the signs at the identity, on $H \setminus \{1\}$, and outside H shows

$$t = \|\mu - q\|_1 = 4p(1 - 1/n), \qquad \delta = 2p[1 - p(1 + 1/n)].$$

Thus $t > 2\delta$ whenever $p > 1/(n+1)$. Take $H = C_n$, choose $p = 2/n$ with $n \geq 16$, and let $n \to \infty$. Both t and δ tend to zero, while the strict inequality holds for every such n.

For distinct subgroups H, K,

$$\|m_H - m_K\|_1 = 2\left(1 - \frac{|H \cap K|}{\max\{|H|, |K|\}}\right) \geq 1.$$

For our choice of p,

$$t + 2\delta = \frac{16}{n} - \frac{24}{n^2} - \frac{16}{n^3} < 1 \qquad (n \geq 16).$$

Therefore $\|\mu - m_K\|_1 \geq 1 - t > 2\delta$ for every $K \neq H$. No other subgroup can remove the counterexample. The group size varies in this argument, as required to rule out a uniform positive threshold. □

A A linear factorization through approximate algebras

For completeness, we record a stronger factorization estimate than the main proof requires. The linear approximate retract statement below appears in Osborne's public manuscript [9, §4, equations (F2)–(F3)]. The proof given here uses the scalar consequences of Kitaev's Definition 2.2 and Theorems 2.3 and 12.1 [10]; it does not assume a general repair theorem for quantum channels.

Lemma A.1 (Approximate stochastic retract)**.** *There are absolute constants $K \geq 1$ and $\eta_* > 0$ such that every stochastic $n \times n$ matrix Q with*

$$\eta = \left\|Q^2 - Q\right\| \leq \eta_*$$

has stochastic factors $A\colon \mathbb{R}^k \to \mathbb{R}^n$ and $M\colon \mathbb{R}^n \to \mathbb{R}^k$, for some $k \geq 1$, satisfying

$$\|AM - Q\| \leq K\eta, \qquad \|MA - I_k\| \leq K\eta. \tag{A.1}$$

Proof. Let P be the projection from Lemma 2.2, so $P\mathbf{1} = \mathbf{1}$ and $d = \|P - Q\| \leq 4\eta$.

To apply Kitaev, let $D\colon M_n(\mathbb{C}) \to \mathbb{C}^n$ extract the diagonal and $J\colon \mathbb{C}^n \to M_n(\mathbb{C})$ insert it. The map $\Phi = JQD$ is unital and completely positive. Moreover

$$\left\|\Phi^2 - \Phi\right\|_{\mathrm{cb}} = \left\|Q^2 - Q\right\| = \eta.$$

Indeed, for a map between finite commutative function algebras, the norm at every matrix level is bounded by the maximum absolute row sum, and scalar sign choices give the reverse bound. Composition with D, J preserves this equality. The repaired map in Kitaev's equation (12.2) is JPD: the repair function has value zero at zero and hence annihilates off-diagonal matrices.

Put $E = \operatorname{im} P \subset \mathbb{C}^n$ and give it product $x \star y = P(xy)$, with pointwise multiplication inside the parentheses. Kitaev's Theorem 12.1 and then Theorem 2.3 provide a finite-dimensional

C^*-algebra B and a bijective, exactly involution-preserving linear map $T\colon B \to E$. After choosing its direction and adjusting an absolute constant, there is $\kappa \le K_1\eta$ such that

$$(1-\kappa)\,\|u\| \le \|Tu\| \le (1+\kappa)\,\|u\|\,, \qquad \|T\mathbf{1}-\mathbf{1}\| \le \kappa, \tag{A.2}$$

$$\|T(uv) - (Tu)\star(Tv)\| \le \kappa\,\|u\|\,\|v\|\,. \tag{A.3}$$

All smallness conditions and K_1 are independent of n.

The product $\star$ is exactly commutative, including when Q comes from convolution on a nonabelian group. Thus

$$(1-\kappa)\,\|uv - vu\| \le 2\kappa\,\|u\|\,\|v\|\,.$$

A noncommutative finite-dimensional C^*-algebra contains a matrix block with two norm-one elements whose commutator has norm two. For $\kappa < 1/2$ this is impossible, so B is $\mathbb{C}^k$. Since T preserves involution, its matrix in the standard real coordinates is real.

Set $S = T^{-1}P$. Then $ST = I_k$, $TS = P$, and

$$\|S\| \le \frac{1+d}{1-\kappa}, \qquad \|S\mathbf{1}-\mathbf{1}\| \le \frac{\kappa}{1-\kappa}.$$

Apply Lemma 2.1 to every row of T, S, with $\kappa \le 1/4$, reducing η_* if necessary. The resulting stochastic A, M satisfy

$$\|A-T\| \le \kappa, \qquad \|M-S\| \le \sigma := \frac{d+\kappa}{1-\kappa}.$$

Using $\|A\| = \|M\| = 1$ and $\|T\| \le 1+\kappa$ gives

$$\|AM - Q\| \le d + \kappa + (1+\kappa)\sigma = \frac{2(d+\kappa)}{1-\kappa},$$

$$\|MA - I_k\| \le \kappa + (1+\kappa)\sigma \le \frac{2(d+\kappa)}{1-\kappa}.$$

Together with (2.2), these estimates prove the assertion after reducing η_*. □

References

[1] Y. Kawada and K. Itô, *On the probability distribution on a compact group. I*, Proceedings of the Physico-Mathematical Society of Japan, Third Series 22 (1940), 977–998; Theorem 3, p. 981. Original article. MR0003462

[2] Ekene E., *Classifying the nature of idempotents on finite groups*, MathOverflow question, 15 November 2025; answer by Freddie Manners, 29 November 2025, edited 2 December 2025. Question and answer.

[3] S. Saeki, *On norms of idempotent measures*, Proceedings of the American Mathematical Society 19 (1968), 600–602. doi:10.1090/S0002-9939-1968-0225102-6. MR0225102

[4] J. Mudge and H. L. Pham, *Idempotents with small norms*, Journal of Functional Analysis 270 (2016), no. 12, 4597–4603. doi:10.1016/j.jfa.2016.02.011. Theorem 1.1(i); arXiv:1510.03535v1 (13 October 2015).

[5] M. Zippin, *The finite dimensional P_λ-spaces with small λ*, Israel Journal of Mathematics 39 (1981), 359–364. doi:10.1007/BF02761680.

[6] M. Zippin, correction to *The finite dimensional P_λ-spaces with small λ*, Israel Journal of Mathematics 48 (1984), 255–256. doi:10.1007/BF02761169.

[7] M. Zippin, *Extension of bounded linear operators*, Handbook of the Geometry of Banach Spaces, vol. 2 (2003), 1703–1741; Theorem 2.3. doi:10.1016/S1874-5849(03)80047-5. Author's text. MR1999607

[8] G. Godefroy, N. J. Kalton, and G. Lancien, *Subspaces of $c_0(\mathbb{N})$ and Lipschitz isomorphisms*, Geometric and Functional Analysis 10 (2000), 798–820; proof of Fact 3.3, p. 809. doi:10.1007/PL00001638. Published full text. MR1791140

[9] T. Osborne, *Almost-Idempotent Stochastic Maps via Kitaev Factorization*, public manuscript, `paper/main.tex`, §4, equations (F2)–(F3), snapshot of 11 August 2026, commit `3c5c0b0900c9823f1ba424bfddeab83777f43f93`. Pinned manuscript.

[10] A. Kitaev, *Almost-idempotent quantum channels and approximate C^*-algebras*, arXiv:2405.02434v2, 11 February 2025 (first version: 3 May 2024); Definition 2.2, Theorems 2.3 and 12.1. `https://arxiv.org/abs/2405.02434v2`.

AI assistance

AI assisted the formulation, literature comparison, proof development, and preparation of this manuscript within the Mathematics Knowledge and Connections Program. The author directed the development and revision and takes responsibility for the content.

Funding

This research received no external funding.